HybridRAG-Eval

Hybrid retrieval, reranking and cited answers,
with a reproducible evaluation benchmark

Project report	Retrieval-augmented generation (RAG) over PDF, DOCX and TXT documents
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Repository	HybridRAG-Eval

22	**47**	**0.936**	**5 / 47**
documents (337 pages)	evaluation questions	best Hit@5 (recursive + rerank)	answers graded (API quota limit)

1. Executive summary

HybridRAG-Eval is a question-answering system over a corpus of 22 documents (10 PDF, 6 DOCX, 6 TXT; 337 pages). It combines **BM25 keyword search** and **FAISS vector search**, merges the two rankings with Reciprocal Rank Fusion, re-scores the candidates with a **cross-encoder reranker**, and asks an LLM (Gemini) to answer only from the retrieved passages while citing the source file and page. A 47-question benchmark measures retrieval quality and compares two chunking strategies across four retrieval configurations.

Key findings

- **Hybrid + Rerank was the best configuration for both chunking strategies:** Hit@5 of 0.915 (fixed) and 0.936 (recursive), MRR@5 of 0.782 and 0.801.
- **The reranker gave the most consistent improvement.** It raised Hit@5 by 3 questions (fixed) and 4 questions (recursive) over plain hybrid retrieval.
- **Recursive chunking was slightly ahead of fixed-size chunking, but the difference is not statistically significant** (one question at Hit@5; sign test p of about 0.80 on the 16 questions where the strategies differed).
- **Answer quality was measured on only 5 questions.** All 5 answers per strategy were judged correct and no citation was invented, but the free Gemini API quota ran out, so the full 47-question answer evaluation could not be completed (Section 6).

Requirements coverage

Requirement	Status	Details
20+ documents (PDF / DOCX / TXT)	**Met**	22 documents: 10 PDF, 6 DOCX, 6 TXT
Hybrid retrieval (keyword + vector)	**Met**	BM25 + FAISS, fused with Reciprocal Rank Fusion
Reranking step	**Met**	Cross-encoder, top 20 candidates to top 5
Evaluation set of 30+ question-answer pairs	**Met**	47 questions with ground truth, source and evidence
Retrieval hit rate	**Met**	Hit@1/3/5, DocHit@5 and MRR on all 47 questions
Answer correctness	**Partial**	LLM judge on the first 5 questions only (free API quota)
Compare 2+ chunking strategies, report which is better and why	**Met**	Fixed vs recursive; no significant difference (Section 5)
Answers cite sources	**Met**	[source: file, page X]; citation accuracy and grounding measured
Deliverables: code, eval set, results table, README	**Met**	Repository, eval_set.json, results tables, README and this report

2. System design

The system has three parts: offline indexing, an online query pipeline and an evaluation harness.

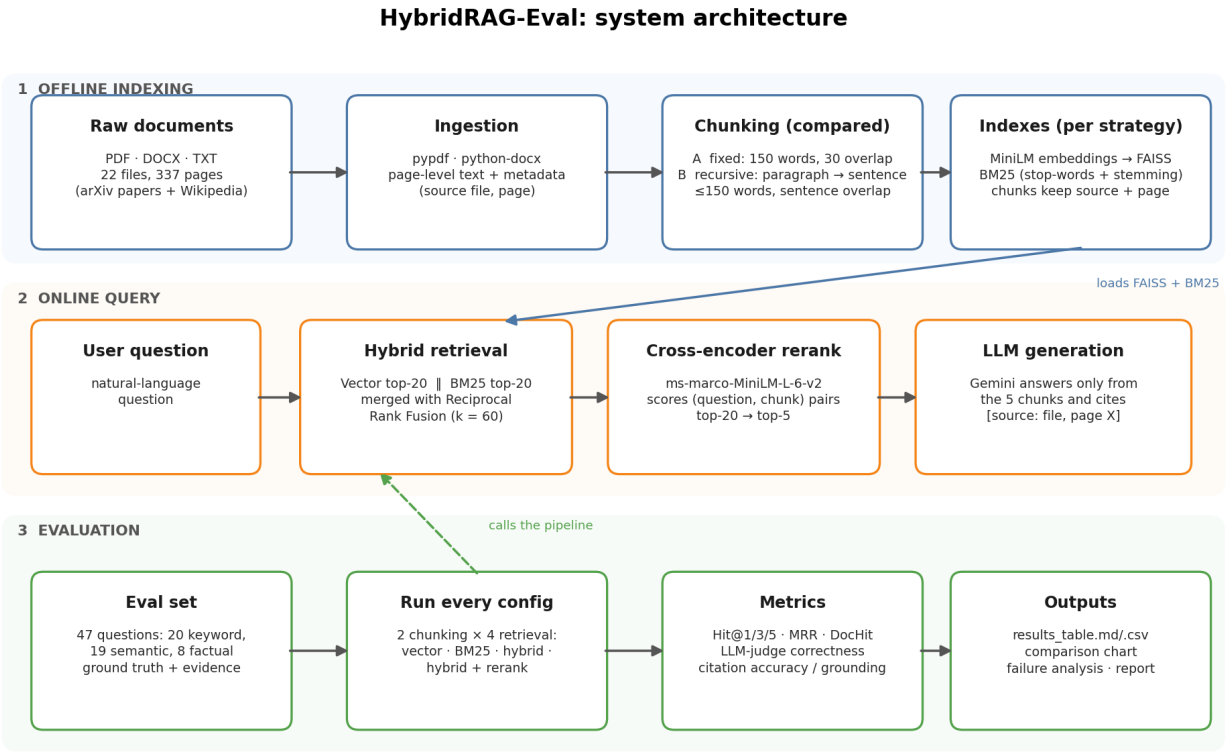


Figure 1. System architecture

- **Ingestion.** PDF, DOCX and TXT files are read page by page, so every chunk keeps its source file and page number. TXT and DOCX files have no pages, so paragraphs are grouped into pseudo-pages of about 3000 characters.
- **Chunking.** Two strategies with the same size budget (up to 150 words) are compared: fixed-size windows and recursive paragraph/sentence chunking.
- **Retrieval.** Vector search (all-MiniLM-L6-v2 embeddings in FAISS) and BM25 each return 20 candidates; Reciprocal Rank Fusion (k = 60) merges them.
- **Reranking.** The cross-encoder ms-marco-MiniLM-L-6-v2 scores each (question, chunk) pair and keeps the best 5.
- **Generation.** Gemini answers only from the 5 chunks, cites [source: file, page X] after each statement and replies "Not found in documents." when the passages do not contain the answer.

3. Corpus and evaluation set

Item	Details
Documents	22 files, 337 pages: 10 arXiv papers (PDF) and 12 Wikipedia articles (6 TXT, 6 DOCX)
Topics	Machine learning and information retrieval: Transformer, BERT, RAG, ResNet, LoRA, Adam, batch normalization, word2vec, dropout, DPR, and general ML articles
Chunks	fixed: 1403 chunks, 136 words on average. recursive: 1403 chunks, 126 words on average
Questions	47 questions: 20 keyword-style, 19 semantic, 8 factual. Each has a ground-truth answer, a source document and evidence phrases

Item	Details
Relevance rule	A retrieved chunk is a hit if it comes from a ground-truth document and contains an evidence phrase. This does not depend on chunk boundaries, so both strategies are scored the same way
Validation	A script checked that each question's source file and evidence phrase exist in the corpus: 0 problems out of 47

Metrics: **Hit@k** (a top-k chunk is relevant), **DocHit@k** (a top-k chunk comes from the right document), **MRR@5** (mean reciprocal rank), **correctness** (LLM judge score of 1, 0.5 or 0), **answer similarity** (embedding cosine similarity to the ground truth), **citation accuracy** (a cited source matches the ground truth) and **citation grounding** (every cited source and page was among the retrieved chunks).

4. Results

4.1 Retrieval quality (47 questions)

Chunking	Retrieval	Hit@1	Hit@3	Hit@5	DocHit@5	MRR@5
fixed	Vector only	0.511	0.787	0.830	0.979	0.644
fixed	BM25 only	0.638	0.787	0.809	1.000	0.718
fixed	Hybrid (RRF)	0.681	0.830	0.851	0.979	0.756
fixed	**Hybrid + Rerank**	0.681	0.872	0.915	0.979	0.782
recursive	Vector only	0.447	0.745	0.851	0.979	0.606
recursive	BM25 only	0.660	0.809	0.830	1.000	0.736
recursive	Hybrid (RRF)	0.638	0.809	0.851	0.979	0.729
recursive	**Hybrid + Rerank**	**0.723**	**0.894**	**0.936**	0.979	**0.801**

One question is worth 0.021 (1/47) on any Hit metric.

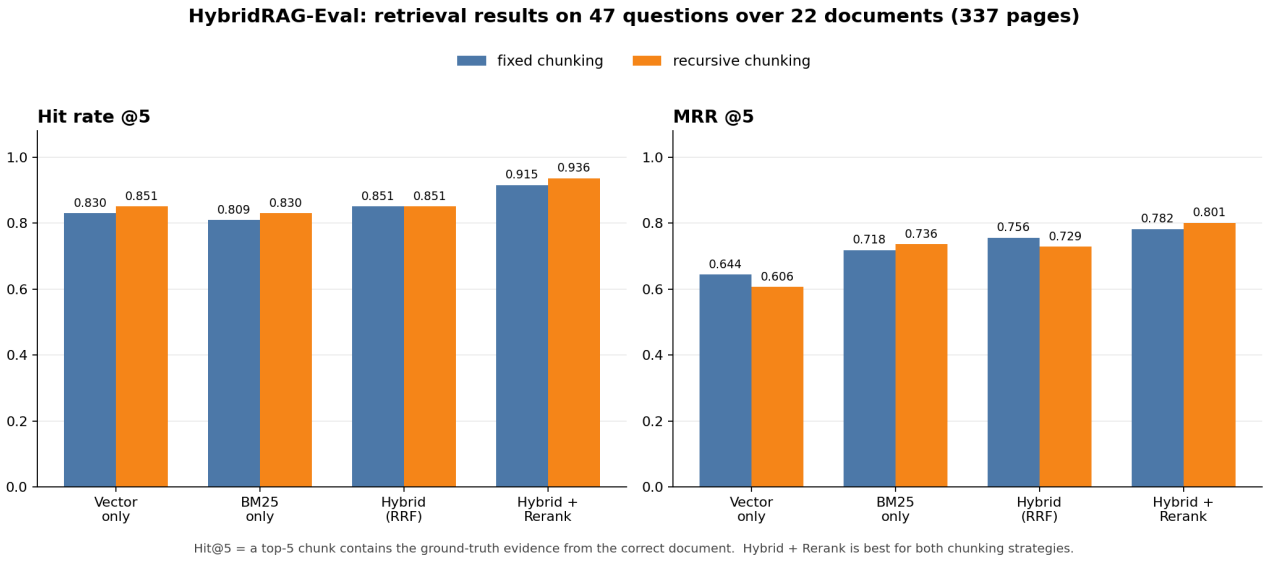


Figure 2. Hit@5 and MRR@5 for all configurations

4.2 Hit@5 by question type (Hybrid + Rerank)

Chunking	Factual (8 questions)	Keyword (20 questions)	Semantic (19 questions)
fixed	0.875 (7/8)	0.950 (19/20)	0.895 (17/19)
recursive	0.875 (7/8)	1.000 (20/20)	0.895 (17/19)

4.3 Answer quality (pilot: first 5 questions only)

Chunking	Graded	Correctness	Answer similarity	Citation accuracy	Citation grounding	Abstained
fixed	5	1.000	0.724	0.800	1.000	0
recursive	5	1.000	0.738	0.800	1.000	0

All 10 pilot answers were judged fully correct and no cited page was invented. In 1 of 5 answers per strategy none of the cited sources matched the ground-truth document (the cause was not investigated). With 5 questions these numbers show that the generation and citation path works; they do not measure answer quality reliably. The same model (Gemini 3.5 Flash) generated and judged the answers.

5. Chunking comparison: which performs better, and why

Result. With the full pipeline, recursive chunking scored higher: Hit@5 0.936 against 0.915, Hit@1 0.723 against 0.681 and MRR@5 0.801 against 0.782. It missed 3 questions at Hit@5 and fixed chunking missed 4.

Is the difference real? Probably not. The Hit@5 gap is a single question. Question by question, the reciprocal rank differed on 16 questions: recursive was better on 9, fixed on 7, and 31 were tied. A sign test on those 16 questions gives p of about 0.80, so there is no evidence that either strategy is better. The fair conclusion is that **recursive chunking was slightly ahead in this experiment, but the two strategies are practically equivalent at this corpus size and chunk size.**

An observation worth testing. Recursive chunking won 6 of the 7 semantic questions on which the strategies differed (Q13, Q16, Q18, Q20, Q37, Q46; fixed won Q11). Fixed chunking won 5 of the 7 keyword questions on which they differed (Q2, Q3, Q9, Q10, Q35; recursive won Q29 and Q42). Of the 2 factual questions that differed, fixed won Q1 and recursive won Q6.

Possible reasons (hypotheses, not verified)

- Recursive chunking keeps sentences and paragraphs whole, so a concept explained over several sentences usually sits in one chunk. That could help semantic questions, where the cross-encoder needs the full explanation to recognise the answer.
- Fixed windows overlap by 30 words and ignore sentence boundaries, so a specific number or name is almost always inside some chunk. That could help keyword questions, where BM25 only needs the term to be present.
- Fixed windows can cut a sentence in half, and a cut chunk may rank lower even though the evidence is present.

These explanations were not tested. A chunk-size ablation, inspection of the 16 differing questions (evaluation/failure_analysis.py) and a larger evaluation set are needed to confirm or reject them.

Retrieval methods

- **Reranking** raised Hit@5 from 0.851 to 0.915 (fixed, +3 questions) and from 0.851 to 0.936 (recursive, +4 questions). Recursive Hit@1 rose from 0.638 to 0.723 (+4 questions).
- **Hybrid fusion alone was not always better than the best single retriever.** With fixed chunks it beat both vector and BM25 on MRR (0.756 against 0.644 and 0.718). With recursive chunks its MRR (0.729) was slightly below BM25 (0.736) and its Hit@5 tied with vector search (0.851).
- **Finding the right document is nearly solved** (DocHit@5 0.979 to 1.000); remaining errors are mostly about picking the right chunk inside the right document.
- **Vector search was weaker than expected on semantic questions** (Hit@5 0.737 against 0.789 for BM25). A small embedding model that truncates input at about 256 tokens, or vocabulary overlap between questions and sources, may explain it; this was not investigated.

6. Issues faced and how they were handled

The table lists every problem met while building and evaluating the system, its cause and how it was handled.

#	Issue	Symptom and cause	Resolution	Status
1	Script and test file not found	`python scripts/download_corpus.py` and `python test_key.py` failed with "No such file or directory" because the folder and files had not been created in the project root.	Created the `scripts` folder and the files in the right place and ran the commands from the project root.	**Resolved**
2	Virtual environment activation (Windows)	Windows PowerShell restricts running scripts by default, which affects the venv activation script.	Activated the environment with a process-scoped execution policy (Set-ExecutionPolicy -Scope Process RemoteSigned).	**Resolved**
3	Missing dependency	Building the index failed with `ModuleNotFoundError: No module named 'sentence_transformers'`; the requirements were not fully installed in the virtual environment.	Installed sentence-transformers, ran pip install -r requirements.txt and verified all imports in one command.	**Resolved**
4	Hugging Face download error	Loading the reranker failed with `OSError: couldn't connect to huggingface.co`. A connectivity test then returned HTTP 200, so the failure was temporary.	Downloaded the model explicitly with a longer timeout (HF_HUB_DOWNLOAD_TIMEOUT=120). The snapshot was about 850 MB because it included alternative model formats; later runs used the local cache.	**Resolved**
5	API key not found	`RuntimeError: GEMINI_API_KEY not found`. The key file did not exist as `.env`; it had been saved as `.env.txt` (and first had to be created from .env.example).	Renamed the file to `.env` in the project root. The test call then returned a reply from the model.	**Resolved**
6	Retired Gemini model	The original plan used gemini-2.5-flash, which had been retired in October 2026, so calls to it would fail.	Switched the default to gemini-3.5-flash and made the model configurable through .env.	**Resolved**
7	Free-tier API quota (429)	`429 RESOURCE_EXHAUSTED` ("You exceeded your current quota") at question 6. Four retries (8, 16, 32 and 64 seconds) did not help and a later attempt failed the same way. A full run needs about 190 LLM calls (47 questions x 2 chunking strategies x answer and judge), far above the free-tier limits.	Added an answer cache with resume, detection of daily-quota errors, and a provider-agnostic LLM layer (Groq, OpenRouter, Ollama, separate judge model). Decision: stop the answer evaluation at 5 questions and document the limitation.	**Open**
8	Result files overwritten	A 5-question trial run replaced the result files of the full 47-question retrieval run.	Re-ran the retrieval evaluation on all 47 questions to regenerate them; failure analysis is run only after the full retrieval run.	**Resolved**
9	No pages in TXT and DOCX	These formats have no pages, but citations need a page number.	Paragraphs are grouped into pseudo-pages of about 3000 characters; for these files "page" means section.	**Mitigated**
10	Harmless warnings	"Unauthenticated requests to the HF Hub" and a Windows symlink warning for the Hugging Face cache.	Notices only (slower downloads and more disk use). Setting HF_TOKEN would raise the download rate limits.	**No action**

Lessons learned

- Run commands from the project root and check file names and extensions (for example `.env` versus `.env.txt`).
- Keep API keys only in `.env`, which is excluded from Git; never put them in chats, screenshots or commits.
- Free-tier LLM quotas are small. Estimate the number of calls before an evaluation, cache every result and design runs that can resume.
- A small evaluation set cannot separate strategies that differ by one or two questions; report the uncertainty.

7. Limitations

- Answer quality covers 5 of 47 questions because of the free API quota; retrieval metrics cover all 47.
- The generator and the judge were the same model in the pilot, which can favour the generated answers.
- The evaluation set was written for this project and was not independently reviewed. Evidence-phrase matching can miss valid passages worded differently.
- With 47 questions, differences of one or two questions (2 to 4 percentage points) are within noise.
- Chunks do not cross page boundaries, so an answer split across two pages can be cut.
- One embedding model, one chunk size and one overlap setting were tested.

8. Conclusion and future work

The full pipeline works end to end. Hybrid retrieval with reranking finds relevant evidence in the top 5 for 91.5% (fixed) and 93.6% (recursive) of the 47 questions, and the pilot answers were correct and cited real retrieved pages. **Reranking is the most valuable component**, and **recursive chunking is slightly ahead but not significantly better than fixed-size chunking** on this corpus.

Next steps

- Complete the answer-quality run on all 47 questions with a higher quota or another provider, using a different model as judge.
- Add bootstrap confidence intervals and a chunk-size ablation (for example 100, 150 and 250 words).
- Inspect the questions where the strategies disagree and enlarge the evaluation set, reviewed by a second person.
- Try a stronger embedding model and expose the pipeline through a FastAPI /ask endpoint.

Appendix A. How to reproduce

```
python -m src.indexing                                  # ingest, chunk and index both strategies
python -m evaluation.validate_eval_set                  # check the evaluation set against the corpus
python -m evaluation.run_evaluation --skip-answers      # retrieval metrics (no LLM calls)
python -m evaluation.run_evaluation --max-questions 5   # retrieval + answer quality on 5 questions
python -m evaluation.failure_analysis                   # misses and chunking differences
python -m src.pipeline -q "How many attention heads does the Transformer use?"
```

Setup: Python 3.10 or 3.11, `pip install -r requirements.txt`, and a `.env` file with the LLM key. Models are downloaded from Hugging Face on first use.